EXP 3:Write the python program for Water Jug Problem

```
from collections import deque

def water_jug():
    queue = deque([((0, 0), [])])
    visited = set()

    while queue:
        (a, b), path = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))

        if a == 2 or b == 2:
            print("Solution:")
            for state in path + [(a, b)]:
                print(state)
            return

        states = [
            (4, b),
            (a, 3),
            (0, b),
            (a, 0),
            (a - min(a, 3 - b), b + min(a, 3 - b)),
            (a + min(b, 4 - a), b - min(b, 4 - a))
        ]

        for new_state in states:
            if new_state not in visited:
                queue.append((new_state, path + [(a, b)]))

water_jug()
```

Output


```
Solution:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
```